

Cairo Real Estate Market Analysis

How Location & Property Type Drive Prices Across 28 Districts in Cairo

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11,418

Raw Listings

8,851

Clean Records

28

Districts

10

Property Types



Introduction

Cairo's housing market is shaped by rapid urbanization and surging demand. This study analyzes 8,851 cleaned property listings scraped from Egyptian real estate platforms across 28 districts and 10 property types, aiming to statistically determine what drives price per square meter.

Research Question: Do location and property type significantly affect price per sqm?



Hypothesis

H₀: Location and property type have NO significant effect on price per sqm.

H₁: At least one factor significantly predicts price per sqm.



Statistical Test Results

Test	Factor	p-val	Result
ANOVA	Property Type vs Price/sqm	<0.05	✓ Reject H ₀
ANOVA	District vs Price/sqm	<0.05	✓ Reject H ₀
Kruskal	Property Type vs Price/sqm	<0.05	✓ Reject H ₀
Kruskal	District vs Price/sqm	<0.05	✓ Reject H ₀

All 4 tests **reject H₀** ($p < 0.05$) — both district and property type are statistically significant predictors of price per sqm.



Methodology

Data: 11,418 Kaggle listings — Egyptian real estate platforms (Spring 2026).

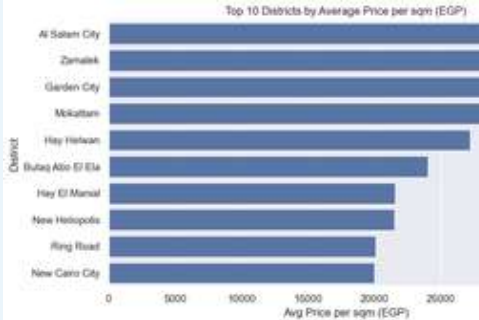
- Removed 412 'Ask' entries (non-numeric prices)
- IQR-based outlier removal on price & size_sqm
- Parsed location → city / district / compound columns
- Engineered features: price_per_sqm, total_rooms

Statistical Tests: One-way ANOVA (parametric) + Kruskal-Wallis (non-parametric) + OLS regression on log-transformed price.



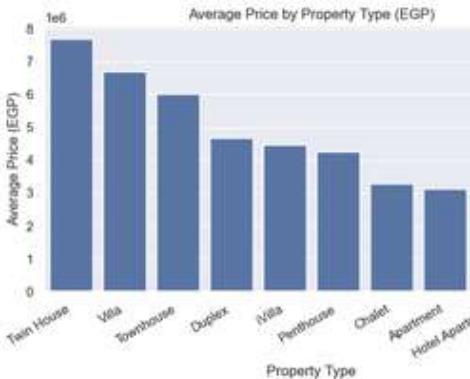
Key Visualizations

Top 10 Districts by Avg Price per sqm



Al Salam City leads at ~34,000 EGP/sqm — surpassing even Zamalek and Garden City. New Cairo averages ~20,000 EGP/sqm, a 70% gap that proves location is the single strongest price driver in Cairo's market.

Average Price by Property Type



Villas and Twin Houses command more than 2× the price of a typical apartment. This stark contrast confirms that property type is a powerful independent predictor — buyers pay a large premium for space and property category beyond square meters alone.

Property Size vs. Price by Type (Log Scale)



Conclusions

Key Results

- H₀ rejected — $p < 0.05$ across all 4 tests
- District & type are both significant
- Location swings prices 50%+ above average
- Villas cost 2× more than apartments



Modelling Insights

- Size alone is NOT a strong price predictor
- Log-transform essential for regression
- Bedrooms & bathrooms correlated ($r = 0.93$)
- Use total_rooms to avoid multicollinearity



Implications

- Target investments in under-priced districts
- Expand apartments to close affordability gap
- Type + location > size for pricing decisions
- ANOVA & Kruskal-Wallis both confirm findings